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Decoding Your Ride:

What Information Builds Human Trust on Real Roads?

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The age of self-driving cars is no longer a distant sci-fi dream; it's an emergent reality. Highly automated vehicles, those capable of handling all driving tasks under specific or all conditions (SAE Level 4 or 5), promise a future where we can reclaim our commute time for work, relaxation, or entertainment. However, as with any technological shift, widespread adoption hinges on public trust and acceptance. For some individuals, the idea of ceding control to a machine, especially a first-time experience, can evoke anxiety [1], particularly when the vehicle behaves in unexpected ways.

This is where the concept of “explanations” becomes relevant. We are discovering that providing passengers with information about what an automated vehicle is “judging” or “perceiving” can enhance their experience. Explanations can give passengers an increased sense of control and help them appropriately calibrate their trust based on the vehicle's actual capabilities [2]. Ultimately, this can reduce anxiety, encouraging automation use and making the journey more comfortable and productive.

THE HUMAN ELEMENT: OUR PASSENGER-IN-THE-LOOP APPROACH TO REAL-WORLD INSIGHTS

While simulations offer valuable insights, we recognized the importance of evaluating in-vehicle explanations under real-world conditions to fully capture passenger responses. Traditional lab-based driving simulators, while offering control over experimental variables, operate in inherently safe settings. These controlled environments may not always reflect the full spectrum of emotional and cognitive responses experienced in real-world scenarios, where subtle stressors and the perception of actual consequences are more pronounced.

By conducting our experiments in an actual vehicle navigating real-world traffic, including navigating a crowded street, we immersed our participants in a more ecologically valid environment. They experienced the unpredictable flow of other cars, the changing road surfaces, the ambient sounds, and the subtle kinesthetic feedback from moving a car, all factors that contribute to a realistic driving experience and can influence perceptions of safety, trust, and anxiety when interacting with automation. This on-road approach allows us to observe and measure

responses in a context where the potential for unexpected events is real, providing a more valid environment to test how explanations affect the passenger experience.

Our method involved transforming a standard vehicle into an on-road automated driving simulator, extending our MAXIM platform [3, 4]. We leveraged a Wizard-of-Oz (WoZ) technique [5], where a human “wizard” driver controlled the vehicle, integrated with an extended reality (XR) system for passengers. The study participant, seated in the passenger seat, wore a Virtual Reality (VR) headset. Inside the VR environment, they experienced the drive as though they were in the driver’s seat of a highly automated vehicle; the actual wizard driver was visually removed from their perception, and a live 360° camera feed of the real-world surroundings was integrated into their virtual view (Figure 1).

This mixed-reality (MR) approach creates the illusion of self-driving while also serving as a tool for interface design and evaluation. Within the VR environment, we projected visual explanations onto a simulated Windshield Display (WSD). This digital overlay onto the live, real-world view enabled rapid testing and refinement of AV communication interfaces within a realistic

context. The approach provides flexibility, allowing instant changes to explanation type, content, and timing without physical modifications to the car itself.

By putting the passenger directly “in the loop” under these naturalistic conditions, we aimed to gather richer, more reliable data on how explanations truly shape the passenger experience – their trust, feelings of safety, and overall acceptance of the technology. Importantly, this on-road Wizard-of-Oz setup, while providing realism, also allowed us to navigate potential ethical and regulatory concerns associated with testing autonomous systems on public roads, ensuring that experimental conditions maintained safety levels comparable to conventional on-road manual driving studies.

PEEKING UNDER THE HOOD: THE EXPLANATIONS WE TESTED

The key interface for delivering information in our study was the simulated WSD, where visual explanations about the vehicle’s understanding of its surroundings were projected. Our study focused on two main types of visual information:

1. Perception Information: This involved showing a “semantic segmentation” map, which highlights and categorizes objects the car detects, such as other vehicles, pedestrians, and cyclists, by overlaying them with a specific color (yellow in our study) on the WSD. The idea was to let passengers see what the car perceives in its environment.

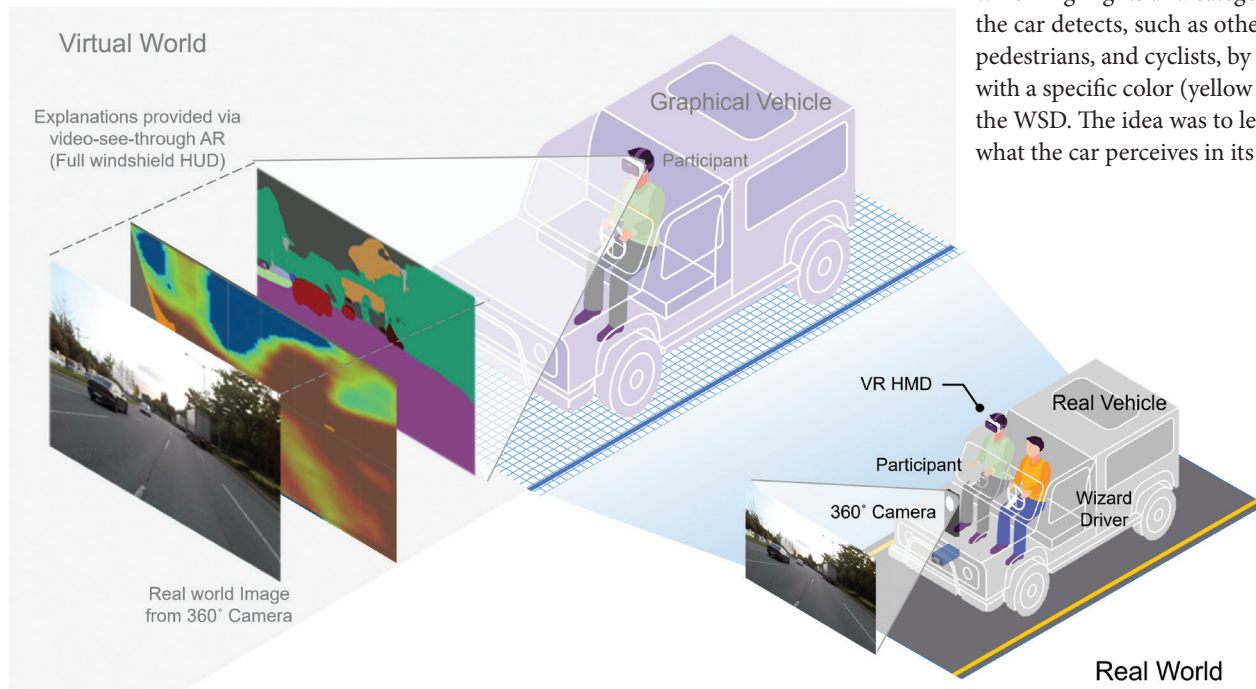


FIGURE 1. Overview of the on-road mixed-reality platform. Real-world driving is captured (360° camera) and presented to the participant (VR HMD), augmented with visual explanations (simulated windshield AR display). The wizard driver is hidden in the virtual view.



2. Attention Information: Using a technique called Grad-CAM [6], our system visualized the areas of the driving scene that the car's AI was paying most attention to when assessing traffic risk. This was displayed as a heatmap where warmer colors indicated regions of higher focus.

We also experimented with *when* these explanations were provided:

- **Continuously:** Information was constantly displayed on the WSD.
- **Risk-Adaptively:** Explanations only appeared when an algorithm we developed (trained on datasets like the Car Crash Dataset [7]) predicted a potentially risky traffic scenario.

A group of 30 participants experienced these different explanation conditions while being driven on a variety of real roads, including urban streets, arterial roads, and local highways (Figure 2). After each drive, they provided feedback through questionnaires on system usability (using scales like the System Usability Scale, SUS), trust, perceived safety, situational awareness (using the Situational Awareness Rating Technique, SART [8]), and cognitive load. Their physiological responses, like heart rate and electrodermal activity [9] (an indicator of arousal), were also monitored.

WHAT TO EXPLAIN? SEEING (THE CAR'S PERCEPTION) IS BELIEVING

Among the explanation types tested, providing perception information (the segmentation map) yielded the most positive results across passenger experience metrics. Participants generally found that seeing what the car "saw" significantly improved:

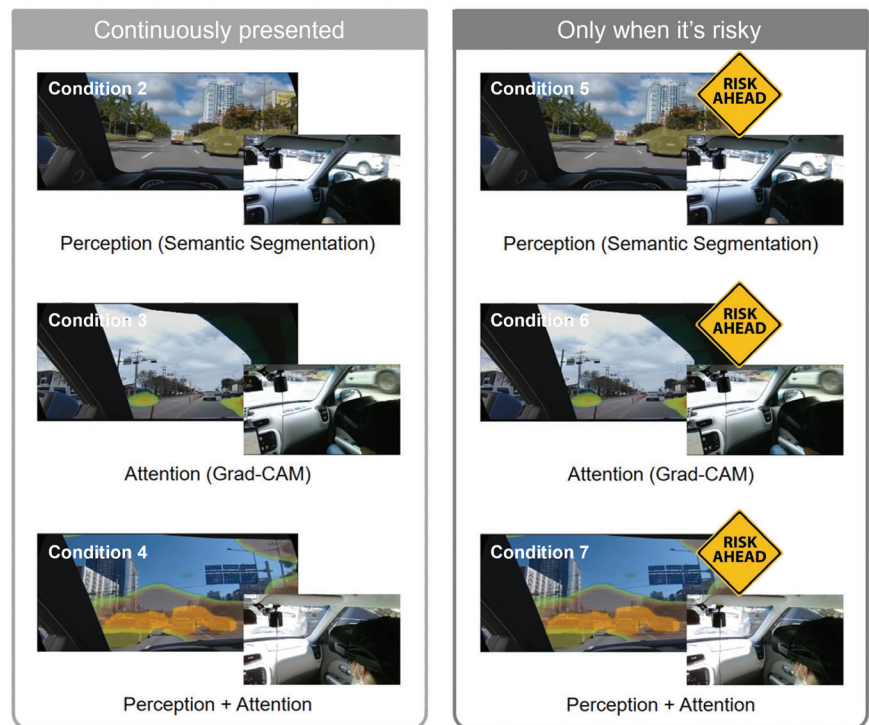


FIGURE 2. Experimental conditions comparing different types of automated vehicle explanations. The study evaluated seven conditions: (1) No Explanation as baseline, continuously presented explanations including (2) Perception with semantic segmentation, (3) Attention using Grad-CAM visualization, and (4) combined Perception + Attention; alongside risk-triggered explanations (5-7) that displayed the same three explanation types but only when the driving situations were perceived as hazardous.

- **Usability:** The system felt easier to use.
- **Trust:** They trusted the automated system more.
- **Perceived Safety:** They felt safer during the ride.
- **Situational Awareness:** They had a better understanding of the driving environment.

Crucially, providing this perception information did so without adding a significant cognitive burden. This aligns with previous laboratory findings suggesting that segmentation visualization helps passengers by enhancing trust and situational awareness [10].

In contrast, the saliency-based attention map (Grad-CAM heatmap) was evaluated poorly with end-users, despite it being considered as one of the most popular ways to decode the intention of Visual AI algorithms. Participants found it:

- **Less Usable and Trustworthy:** It scored lower on usability and trust compared to perception information.

- **More Cognitively Demanding:** It actually increased the perceived cognitive load. Participants mentioned it was difficult to understand and required interpretation – they had to figure out why the car was focusing on certain areas, a task made difficult by the rapidly changing driving scene.

Our study suggests that for explanations to be effective, they need to be direct and easily understood, especially in cars on roads where the environments change rapidly and the vehicle reacts accordingly, which already is a lot of information to process. Simply showing where an AI is "looking" isn't enough if the passenger can't quickly grasp the implication.

WHEN TO EXPLAIN? THE CASE FOR ADAPTIVE TIMING

The question of *when* to provide explanations yielded more nuanced findings. While the continuous display of perception information was generally well-received, we explored

whether providing explanations only during “risky” moments could be beneficial.

We found that **risk-adaptive explanations** could be effective when passengers might otherwise be overwhelmed by too much information. This effect was most pronounced when both perception and attention information were combined. Providing this wealth of information continuously led to the highest cognitive load and was also reflected in increased physiological arousal (phasic EDA), an indicator sensitive to cognitive load [11]. However, when this combined information was shown *only if risky*, the cognitive load and arousal were lower, and trust and perceived safety actually improved compared to showing just attention information. This suggests that an adaptive timing strategy can make complex information more digestible in vehicles.

Interestingly, risk-adaptive explanations also contributed to higher overall situational awareness, even though less total information was delivered. Some participants appreciated getting information selectively when it was most relevant.

However, this adaptive timing wasn’t a perfect solution. A key challenge arose when the car’s algorithmic assessment of risk didn’t match the passenger’s own perception. If the system flagged a situation as risky when the passenger didn’t agree or failed to provide an explanation when the passenger felt endangered, it could lead to frustration or reduced trust. This suggests the need for algorithms that not only detect risk but also align with human intuition about what constitutes a moment requiring explanation.

BUILDING THE BRIDGE TO ACCEPTANCE

Ultimately, the goal of these explanations is to foster passenger acceptance of automated vehicles. Our study modeled how different factors interrelate and found that explanations primarily boost acceptance by enhancing

understandability, perceived safety, and trust. When passengers understand what the car is doing and why, and feel safe and confident in its abilities, they are more willing to use the technology, feel more capable of using it (self-efficacy), and develop a more positive attitude towards it, aligning with frameworks like the Autonomous Vehicle Acceptance Model [1] and general theories of trust in automation [12].

INTERPRETING IMPERFECTIONS: PASSENGER RESPONSES TO EXPLANATION ERRORS

Our research also shed light on how passengers perceive errors in explanations, which is critical for building robust trust.

- What to Explain (Perception Errors):** Participants were more forgiving of “false positives” – the car mistakenly highlighting things like roads or trees as objects – than “false negatives” – the car failing to perceive an actual vehicle or pedestrian. If the car missed something important, it dented their confidence. It’s also crucial for passengers to understand the limits of the explanation system; for instance, some participants noted the car did not show information about lateral observations, such as looking left or right, which was outside the scope of the forward-facing WSD we used in this study.
- When to Explain (Timing Errors):** Here, the preference seemed to be the opposite. Passengers were more concerned by “false positives” – the car providing a risk explanation when they didn’t feel the situation was risky – than “false negatives” – the car not providing an explanation in a situation they perceived as dangerous. If no explanation appeared during a tense moment, they often assumed the car was handling it safely and didn’t perceive it as a hazard.

Understanding these different reactions to errors guides the design of explanation systems, as these distinct responses to “what” and “when” errors offer considerations for tailoring in-vehicle explanations to be more resilient to imperfections and manage passenger expectations.

THE ROAD AHEAD: FUTURE DIRECTIONS

The VR headset, despite enabling our mixed-reality setup, had field-of-view and framerate limitations, preventing tests with complex non-driving tasks like eating or reading. Future work could explore actual WSDs and various non-driving activities. “Attention” explanations also need refinement with more sophisticated saliency maps (e.g., from AI models for end-to-end driving tasks rather than just risk prediction). Alternative visualizations beyond WSDs, like tablet displays, also warrant investigation.

Furthermore, the binary “risky” vs. “not risky” classification for timing explanations could be made more nuanced. Future work might incorporate multiple risk levels/types or use driver interruptibility research to pinpoint ideal delivery moments. Complementing this, passenger demand for explanations could be directly captured via on-road experiments, considering factors like driving context and passenger state to enable more opportune delivery [13].

Our platform could also be extended to study interactions in vehicles with “parallel autonomy,” where the AI acts as a “guardian angel.” This approach to autonomy focuses not on removing driving responsibilities but on augmenting the driver’s skills as needed, aiming to build safe and reliable systems and potentially enabling driving for a wider range of individuals [14].

CONCLUSION: CLARITY IS KEY ON THE JOURNEY TO AUTONOMY

As highly automated vehicles continue their progression towards wider adoption, fostering passenger comfort, safety, and a sense of control becomes a key enabler for this transition. Our work demonstrates that what automated vehicles explain, and when they explain it, matters immensely.

Our on-road evaluation confirms that clear, direct information about the vehicle’s perception of its environment – essentially letting passengers “see what the car sees”

AS HIGHLY AUTOMATED VEHICLES CONTINUE THEIR PROGRESSION TOWARDS WIDER ADOPTION, FOSTERING PASSENGER COMFORT, SAFETY, AND A SENSE OF CONTROL BECOMES A KEY ENABLER FOR THIS TRANSITION

– is a highly effective way to build trust, improve situational awareness, and enhance the overall user experience without causing information overload [15]. While showing the AI’s “attention” directly via heatmaps was evaluated less favorably by passengers in our study, the concept of adapting explanation timing based on traffic risk shows promise, especially for managing more complex information displays.

The journey to fully embraced autonomous driving is not just about perfecting the technology of driving; it’s also about perfecting the art of communication between human and machine. By designing explanations that are intuitive, timely, and address the nuances of human perception and trust, we can pave the way for a future in which self-driving cars are not just a convenience, but a trusted partner on the road. ■

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